CORAL: Learning Amyloid Fibril Ligand Docking with Cooperative Binding Rewards

Yasheng Sun[1*†] Bohan Li[2*] Youqi Tao[3†] Jürgen Schmidhuber[1]

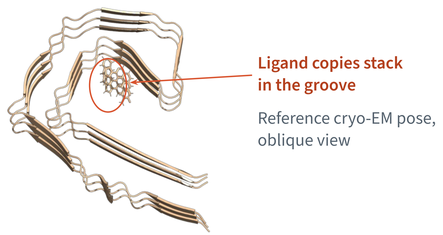
[1]Center of Excellence for Generative AI, KAUST [2]Department of Biomedical Engineering, National University of Singapore [3]Stern Laboratory, Brigham and Women's Hospital, Harvard University [*]Equal contribution [†]Corresponding authors

NeurIPS 2026
Sydney

Project page

1 Cooperative binding

Amyloid fibrils are stacks of identical β-strand layers, ~4.8 Å apart. Ligands intercalate into a cross-β groove and their copies stack along the fibril axis.

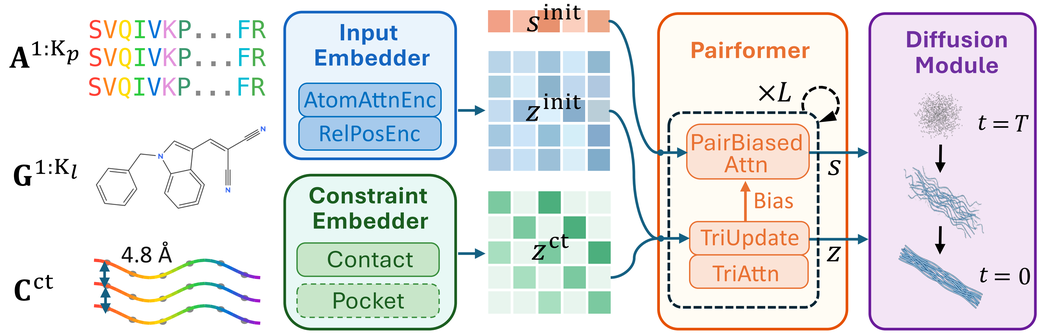


Why fibril docking is difficult

1. **Scarce structures.** Only a handful of fibril–ligand complexes have been resolved.
2. **Cooperative geometry.** Single-pocket models miss periodic multi-ligand stacking.

2 CORAL: fibril–ligand co-folding

Docking as co-folding: one Pairformer–Diffusion model generates all peptide chains and ligand copies together, conditioned on fibril periodicity and the binding pocket.



Inputs: K_p peptide chains, K_l ligand copies, inter-chain Cα contacts

ConstraintEmbedder: ~4.8 Å contact and pocket priors enter the pair track

Pairformer + diffusion: all chains and ligand copies are denoised jointly

3 Docking accuracy

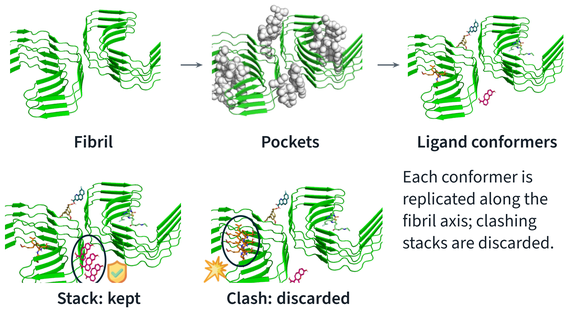
AmyDock-CryoEM Hit@5
50.0%
best baseline **5.6%**
median RMSD **4.89 Å**

AmyDock-Synthetic Hit@5
68.4%
best baseline **3.5%**
median RMSD **3.84 Å**

	AmyDock-CryoEM		AmyDock-Synthetic	
Method	RMSD ↓	Hit@5 ↑	RMSD ↓	Hit@5 ↑
DiffDock	13.68	0.0%	19.68	3.5%
DynamicBind	29.02	0.0%	17.93	1.8%
FABind	24.90	0.0%	26.87	0.0%
NeuralPLexer	19.00	5.6%	16.56	0.0%
FlowDock	21.52	0.0%	19.88	0.0%
Boltz-2	28.67	0.0%	33.87	0.0%
Chai-1	19.92	0.0%	16.52	0.0%
CORAL	**4.89**	**50.0%**	**3.84**	**68.4%**

RMSD: median heavy-atom ligand RMSD of the best pose (Å). Hit@5: fraction with RMSD ≤ 5 Å.

4 Curated stacking data



Each conformer is replicated along the fibril axis; clashing stacks are discarded.

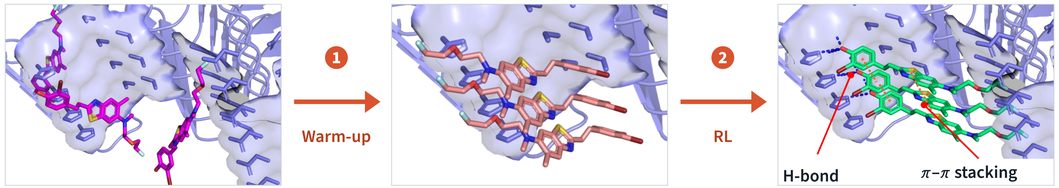
Molecular validity → Computed binding score → Expert review

≈2,000 **Warm-up training poses**
curated synthetic stacks

18 **AmyDock-CryoEM test complexes**
PDB-deposited cryo-EM models

57 **AmyDock-Synthetic test pairs**
expert-curated computational poses

5 Two-stage training



Unaligned ligand copies → Periodic stacking → Tighter contacts

Illustrative example of the binding mode each stage teaches

1 Geometric warm-up

Supervised on curated stacking poses.

$$\mathcal{L}_{\text{SFT}} = \alpha_{\text{diff}}\,\mathcal{L}_{\text{diff}} + \alpha_{\text{dist}}\,\mathcal{L}_{\text{dist}} + \alpha_{\text{conf}}\,\mathcal{L}_{\text{conf}}$$

$\mathcal{L}_{\text{diff}}$: denoise atomic coordinates
$\mathcal{L}_{\text{dist}}$: distogram of pairwise distances
$\mathcal{L}_{\text{conf}}$: pLDDT, PDE and PAE confidence heads

Initialises the RL policy; kept frozen as v_{ref}.

2 Cooperative RL with DiffusionNFT

Fine-tune on sampled poses, weighted by reward.

$$\mathcal{L}_{\text{RL}} = \mathcal{L}_{\text{NFT}} + \beta_{\text{KL}}\,\mathbb{E}\,\|v_\theta - v_{\text{ref}}\|_2^2$$
$$\mathcal{L}_{\text{NFT}} = \mathbb{E}\big[\, r\,\|v_\theta^+ - \mathbf{v}\|_2^2 + (1-r)\,\|v_\theta^- - \mathbf{v}\|_2^2 \,\big]$$

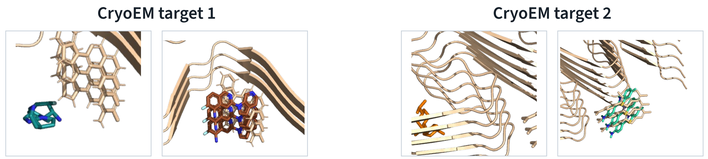
$v_\theta^\pm = (1 \mp \beta)\, v_{\text{old}} \pm \beta\, v_\theta$; v_{old}: sampling policy; $\mathbf{v}$: target velocity
$r \in [0, 1]$: group-normalised reward
v_{ref}: frozen warm-up model, a KL-style velocity anchor

Cooperative binding reward

Protein–ligand score (AutoDock Vina) + **Ligand–ligand stacking** (MMFF94, symmetry neighbours) + **Symmetry consistency** (4.8 Å rigid shift)

normalised per group to $r \in [0, 1]$

6 Binding poses and ablations



Tan ligand: reference; coloured ligand: prediction. Original views; scales differ.

Ablations on AmyDock-Synthetic

Variant (medians)	RMSD (Å) ↓	ΔG_{eff} (kcal/mol) ↓
CORAL (full)	**3.84**	**−2.44**
No stacking term	4.46	−0.95
No symmetry term	4.31	−2.21
No stability regulariser	8.13	−1.45
Warm-up only (no RL)	5.18	−1.58
RL only (no warm-up)	9.27	−0.88

ΔG_{eff} is a computed binding score, not a measured affinity.

Remaining errors: ligands that fold back on themselves, and ligands at the correct site with a sideways orientation.

Supported by a gift from Google, administered through the KAUST Center of Excellence for Generative AI (award 5940).

sunyasheng.github.io/projects/CORAL.html